

My research interest lies in pushing the boundaries of computer systems to improve performance across various domains. I was first drawn to this direction after attending *Systems Savvy: A Woman's Guide to a Career in Computer Systems/Architecture*, where faculty members showcased how thoughtful system-level design underpins high-performance computing. Since then, I have sought out technical challenges that demand both low-level understanding and cross-domain thinking, and I am eager to bring this perspective to cutting-edge research that bridges system architecture and real-world impact.

To act on my interest in systems, I joined an ongoing research project supervised by Professor Qizhen Zhang, where I work directly with the PostgreSQL codebase to analyze and improve database communication efficiency. My work involves tracing client-server interaction pathways and categorizing internal functions to design a cleaner abstraction layer for protocol-level communication, laying the groundwork for future hardware offloading. Through this process, I have developed a deep familiarity with the internals of PostgreSQL, and we are actively investigating how system bottlenecks propagate through database layers. To complement this, I independently studied storage management through CSC443 lecture recordings by Professor Niv Dayan, giving me a detailed view of modern DBMS design. These experiences have equipped me to tackle systems-level challenges that demand both architectural clarity and implementation precision – capabilities I am eager to apply to research like OrcaDB at the OrcaLab.

My interest in pushing system performance also extends to the field of deep learning, where growing model complexity has made efficiency a critical concern. I'm particularly drawn to the challenge of accelerating training and inference through systems-aware techniques, whether by rethinking low-level compiler optimizations for model development and deployment, or by applying high-level methods like quantization and pruning to reduce memory and compute demands. To deepen my practical understanding of GPU-based acceleration, I joined a reading group called *GPU Mode*, where researchers and industry engineers lead discussions on GPU architecture, memory hierarchies, and optimization strategies. These sessions, especially those on profiling and hardware-aware scheduling, have sharpened my ability to think critically about system bottlenecks in large-scale model workloads and motivated my interests in research projects led individually by Professor Maryam Mehri Dehnavi and Professor Nandita Vijaykumar.

I look forward to contributing to research in computer systems and applying my knowledge to accelerate performance in real-world applications. Thank you for considering my application.